

Ex. No.: 1 SIMULATION OF PROCESS SCHEDULING Date:
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AIM

To simulate CPU process scheduling and calculate waiting time and turnaround time.

PROCEDURE

1. Enter the number of processes and burst times.
2. Apply First Come First Serve scheduling.
3. Calculate waiting time for each process.
4. Calculate turnaround time for each process.
5. Display the results and averages.

PROGRAM

```
n = int(input("Enter number of processes: ")) bt = [] wt = [0] * n tat = [0] * n
for i in range(n): bt.append(int(input(f"Enter burst time for P{i+1}: ")))
for i in range(1, n):
    wt[i] = wt[i-1] + bt[i-1]
avg_wt = 0 avg_tat = 0
for i in range(n):
    tat[i] = wt[i] + bt[i] avg_wt += wt[i] avg_tat += tat[i]
    print(f"P{i+1}\tBT={bt[i]}\tWT={wt[i]}\tTAT={tat[i]}")

print("Average Waiting Time =", avg_wt / n)
print("Average Turnaround Time =", avg_tat / n)
```

OUTPUT

Practice 1 - Sample Output

```
python output
Enter number of processes: 3
Enter burst time for P1: 4
Enter burst time for P2: 6
Enter burst time for P3: 2

P1    BT=4    WT=0    TAT=4
P2    BT=6    WT=4    TAT=10
P3    BT=2    WT=10   TAT=12

Average Waiting Time = 4.67
Average Turnaround Time = 8.67
```

RESULT

Thus, the given programs were implemented successfully.

The expected outputs were obtained and the required concepts were verified.

